

Smart Greenhouse Environment Monitoring and Control System

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CERTIFICATE

This is to certify that the project titled "**Greenhouse Monitoring and Control System**" has been successfully completed by the following students as a part of the **Project-Based Learning course (0701230605)** in the **VI Semester** of the **B.Tech Electronics and Telecommunication Engineering** program under my guidance.

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Abstract:

The crop productivity in green houses requires that the environment is kept at optimum levels and this is hard to do with manual methods of monitoring. Manual monitoring of these parameters is time-consuming and often leads to inconsistent environmental control. To address this issue, this project presents an IoT-based smart greenhouse monitoring and control system designed to automate the measurement and regulation of key environmental factors.

The proposed system continuously monitors parameters such as temperature, humidity, soil moisture, light intensity, and carbon dioxide (CO₂) concentration using multiple sensors connected to an ESP32 microcontroller forming the sensor node. The collected sensor data is transmitted wirelessly over long distances using the SX1278 LoRa communication module to a Raspberry Pi gateway. The Raspberry Pi processes the received data and sends it to the Supabase cloud platform using HTTP-based REST APIs for storage and visualization. A user-friendly dashboard designed using Figma and web technologies enables real-time monitoring of environmental conditions and remote control of actuators such as ventilation fans, irrigation pumps, and lighting systems.

The implementation of deep sleep mode in ESP32 enhances energy efficiency by reducing power consumption during idle periods. The developed system demonstrates reliable long-range communication, real-time monitoring, and automated environmental control.

Overall, the proposed solution provides a low-cost, scalable, and energy-efficient smart greenhouse system that improves plant growth conditions, optimizes resource utilization, and supports modern precision agriculture applications.

I: Introduction

1.1 Background

Agriculture plays an important role in ensuring food security, and greenhouse farming has emerged as an effective technique for controlled crop production. A greenhouse provides a semi-controlled environment where critical environmental parameters such as temperature, humidity, soil moisture, light intensity, and carbon dioxide (CO₂) concentration can be maintained at optimal levels to enhance crop growth and productivity. Maintaining these environmental parameters within desired limits is essential for achieving higher crop yield and efficient utilization of resources such as water and energy.

In recent years, advancements in the Internet of Things (IoT) have enabled the development of smart agricultural systems that allow continuous monitoring and automatic control of environmental conditions. IoT-based greenhouse systems utilize sensors, microcontrollers, and wireless communication technologies to monitor environmental parameters in real time and control actuators without human intervention. These systems improve reliability, reduce operational costs, minimize resource wastage, and support modern precision agriculture practices.

2.2 Problem Statement

Conventional greenhouse systems mainly rely on manual monitoring and control methods. In such systems, farmers frequently observe environmental conditions and manually operate ventilation, irrigation, and lighting mechanisms. This process is labour-intensive, time-consuming, and prone to human error. Moreover, environmental conditions inside a greenhouse change continuously throughout the day, making it difficult to maintain optimal conditions using manual methods alone. As a result, crop productivity, water efficiency, and energy utilization are often negatively affected.

Additionally, many existing low-cost greenhouse monitoring systems focus only on temperature and humidity while neglecting other important parameters such as CO₂ concentration. Wireless technologies like Wi-Fi, commonly used in such systems, are not always suitable for large agricultural areas due to their limited communication range and higher power consumption. Therefore, there is a need for a low-cost, long-range, and automated greenhouse monitoring and control system.

2.3 Objectives

The main objectives of the proposed project are:

- To design and develop an IoT-based greenhouse monitoring system for measuring environmental parameters.
- To monitor important parameters such as temperature, humidity, soil moisture, light intensity, and CO₂ concentration.
- To implement long-range wireless communication using the SX1278 LoRa module.
- To use a Raspberry Pi gateway for data processing and cloud communication.
- To develop a cloud-based dashboard for real-time visualization of environmental data.
- To automatically control actuators such as ventilation fans, irrigation pumps, and lighting systems based on predefined threshold values.

2.4 Scope of the Project

The scope of this project includes the development of a smart greenhouse monitoring and control system capable of continuously measuring environmental parameters and transmitting data wirelessly over long distances. The system utilizes ESP32 microcontrollers for sensor data acquisition and LoRa communication for reliable long-range wireless transmission between the sensor node and Raspberry Pi gateway. The gateway processes the received data and forwards it to the cloud for real-time visualization through a dashboard interface.

The project focuses on implementing real-time monitoring, automatic control, and remote visualization of greenhouse conditions. It demonstrates the feasibility of integrating IoT technologies with greenhouse farming to improve crop growth, optimize resource utilization, and support sustainable agricultural practices. The system is designed to be scalable and adaptable for small-scale as well as large-scale greenhouse applications.

2.5 Block Diagram:

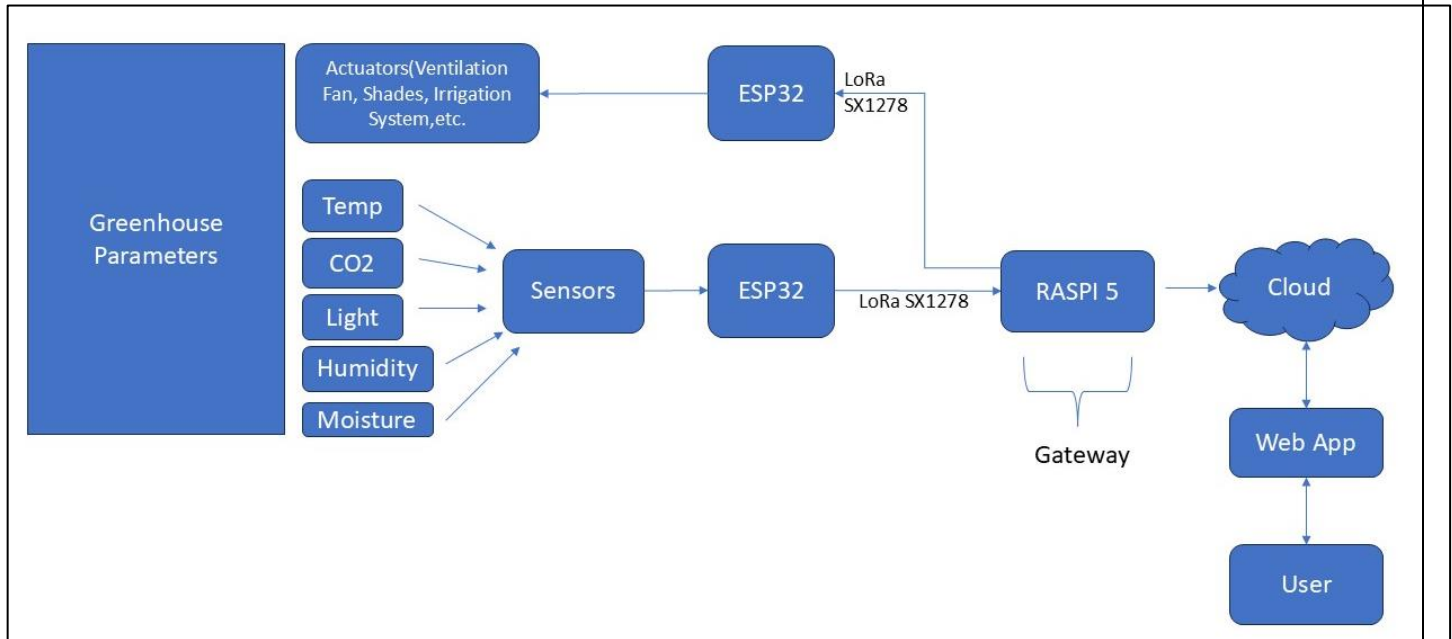


Fig: Block Diagram

II Project details:

2.1 Literature Review:

During the initial phase of the project, an extensive literature review was conducted on IoT-based greenhouse monitoring systems, wireless communication technologies, and smart agriculture solutions. Several research papers related to wireless sensor networks, LoRa communication, cloud-based monitoring, and automated greenhouse control were studied to understand existing technologies and their limitations. Most of the existing greenhouse monitoring systems focus on monitoring basic environmental parameters such as temperature and humidity. However, many systems lack monitoring of important parameters such as carbon dioxide (CO₂) concentration and long-range wireless communication capabilities. In addition,

many conventional systems rely on Wi-Fi communication, which suffers from limited communication range and higher power consumption, making it less suitable for large agricultural areas.

Based on the literature review, several limitations and gaps were identified in existing greenhouse monitoring systems: Most low-cost greenhouse systems focus mainly on temperature and humidity monitoring, while CO₂ concentration monitoring is often neglected. Many systems rely on Wi-Fi communication, which suffers from limited coverage range and high-power consumption, making it unsuitable for large agricultural areas. Several existing systems provide monitoring capabilities but lack automated actuator control mechanisms. Some research focuses only on communication technologies without integrating complete end-to-end greenhouse monitoring systems. Limited emphasis is given to long-range, low-power wireless communication, which is essential for remote greenhouse environments.

The proposed system adopts a gateway-based LoRa communication architecture to overcome limitations of traditional wireless technologies. LoRa technology was selected due to its long communication range, low power consumption, and high reliability, making it suitable for agricultural environments. The system also integrates CO₂ monitoring, automated actuator control, and cloud-based visualization, providing a more complete and efficient greenhouse monitoring solution compared to existing systems.

2.2 System Architecture Design:

The proposed smart greenhouse monitoring system was developed using a combination of hardware and software tools. The hardware components include ESP32 microcontrollers, SX1278 LoRa communication modules, Raspberry Pi gateway, and environmental sensors such as DHT11 (temperature and humidity sensor), BH1750 (light intensity sensor), soil moisture sensor, and SCD40 CO₂ sensor. Actuators such as ventilation fan, irrigation pump, and lighting system were used to control greenhouse conditions.

Based on the literature review findings, the overall system architecture was finalized to ensure reliable monitoring and control of greenhouse environmental conditions. The system consists of a sensor node, LoRa communication module, Raspberry Pi gateway, cloud platform, dashboard interface, and actuator node.

The sensor node is built using an ESP32 microcontroller connected to multiple environmental sensors. The ESP32 collects data from sensors measuring temperature, humidity, soil moisture, light intensity, and CO₂ concentration. After acquiring the sensor data, the ESP32 processes the values and prepares them for transmission.

The collected data is transmitted wirelessly using the SX1278 LoRa module, which enables long-range communication with low power consumption. The transmitted data is received by the Raspberry Pi gateway, which acts as an intermediate processing unit. The Raspberry Pi processes the received sensor readings and forwards the data to the cloud platform for storage and visualization.

The system also supports both automatic and manual control modes. In automatic mode, the Raspberry Pi compares sensor values with predefined threshold levels and generates control commands to maintain optimal greenhouse conditions. In manual mode, users can control devices remotely through the web or mobile dashboard, allowing flexible and efficient greenhouse management.

2.3 Working of System

The working of the proposed system begins with environmental sensors continuously measuring greenhouse parameters such as temperature, humidity, soil moisture, light intensity, and CO₂ concentration. These sensors are connected to the ESP32 microcontroller, which reads the sensor values and processes the collected data. The processed data is then transmitted wirelessly to the Raspberry Pi gateway using LoRa communication, which enables long-range and low-power data transmission.

After receiving the sensor data, the Raspberry Pi processes the information and sends it to the cloud database using HTTP-based REST APIs. The stored data is then displayed on the dashboard interface, allowing users to monitor greenhouse conditions in real time from remote locations.

Based on predefined threshold values, the system automatically generates control commands to maintain optimal environmental conditions inside the greenhouse. These commands are used to operate actuators such as ventilation fans, irrigation pumps, and lighting systems. In addition to automatic control, the system also supports manual control through the dashboard, allowing users to remotely control devices when required.

This system architecture ensures reliable long-range communication, real-time monitoring, automatic environmental control, and efficient resource management, making it suitable for modern smart greenhouse applications.

2.4 Hardware Implementation and Sensor Node Operation:

There has been significant progress in the hardware implementation and testing of the system. In the completed phase, multiple environmental sensors were successfully interfaced with the ESP32 microcontroller to form a reliable sensor node. The BH1750 light intensity sensor was interfaced using the I²C communication protocol, while the soil moisture sensor was connected to the analog input pins of the ESP32 microcontroller. Additionally, the DHT11 temperature and humidity sensor was interfaced with the ESP32 using digital GPIO pins to measure temperature and humidity levels. Proper wiring, calibration of sensors, and stable power supply connections were ensured to achieve accurate and consistent sensor readings.

The ESP32-based sensor node continuously monitors environmental parameters such as temperature, humidity, light intensity, and soil moisture, and processes the values before transmission. The acquired sensor data was found to be stable and reliable, confirming successful hardware integration. The collected data is transmitted wirelessly using the SX1278 LoRa module connected to the ESP32 microcontroller. The transmitted data was successfully received at the Raspberry Pi gateway without packet loss, confirming stable point-to-point LoRa communication.

The sensor node is now functioning reliably, and communication between the ESP32 sensor node and the Raspberry Pi gateway has been fully stabilized. The RSSI (Received Signal Strength Indicator) values were monitored during testing to verify signal strength, and optimal antenna positioning and communication parameters were configured to ensure efficient and reliable long-range data transmission.

To optimize power consumption, deep sleep mode has been implemented on the ESP32 microcontroller. Since continuous monitoring is not always necessary, the ESP32 is programmed to wake up at defined intervals, collect sensor readings, transmit the data over LoRa, and immediately return to deep sleep. This event-driven approach significantly reduces the active runtime of the microcontroller, which in turn extends battery life considerably. Deep sleep mode on the ESP32 draws only a few microamps of current compared to hundreds of milliamps during normal operation, making it well suited for a battery-powered deployment like a greenhouse where frequent charging or replacement is not practical. This implementation ensures the system can run for extended periods autonomously without compromising on data collection reliability.

2.5 Cloud Integration and Dashboard Development:

The dashboard designed to provide centralized interface for monitoring greenhouse conditions and controlling actuators developed with help of Figma to create a responsive and user-friendly layout. The dashboard displays real-time values of environmental parameters such as temperature, humidity, light intensity, soil moisture and CO_2 levels. Each parameters displayed using graphical indicators and numerical values along with status labels High, Low, or Normal values. The dashboard also includes warning notifications highlighting parameters are outside the optimal ranges. Data visualization is implemented using graphical charts that displays the trends of soil moisture, light intensity, temperature and humidity, over time along with minimum, maximum and average values. The interface includes the navigation sections such as Dashboard, Analytics, Controls and Alerts. This integrated dashboard enables efficient monitoring, analysis and management of greenhouse environmental conditions.

Cloud integration implemented in greenhouse monitoring system to enable remote access, storage, monitoring and control of environmental data. The system uses Supabase as a cloud platform which provides real-time database and REST API support. Sensor data is collected from ESP32 and transmitted to Raspberry Pi through LoRa communication. The Raspberry Pi act as gateway sends the received data to cloud database using HTTP POST request. The Raspberry Pi continuously checks cloud database using HTTP GET request and forwards the commands to ESP32 through LoRa communication. The ESP32 then activates the actuators. This cloud-based communication enables real-time monitoring, remote control and storage of environmental data for efficient greenhouse management.

2.6 Greenhouse Architecture and Design:

The greenhouse model design was finalized with approximate dimensions of $20 \times 15 \times 15$ inches, providing sufficient space for the placement of sensors, actuators, and wiring. The model structure includes provisions for installing a ventilation fan, irrigation system, lighting system, and sensor mounting framework. The internal layout was carefully planned to ensure proper positioning of components and easy access for maintenance and wiring management. Initially, a cage-like structure was considered for the greenhouse model design. However, during the planning phase, it was observed that installing sensors, actuators, and wiring inside the structure was inconvenient and difficult to manage. Due to these practical limitations, the dome-shaped design plan was discontinued.

The final greenhouse model was successfully constructed using acrylic sheets, which provide a strong, lightweight, and transparent enclosure suitable for greenhouse applications. Acrylic sheets allowed easy mounting of sensors and actuators while ensuring clear visibility of the internal components. This design choice improved structural stability, simplified installation, and supported efficient arrangement of all

greenhouse components. The installation of sensors, actuators, and wiring within the acrylic enclosure has been completed, and the greenhouse model is now fully prepared for system operation and testing.

2.7 Actuator Node Operation:

The actuator node consists of a second ESP32 microcontroller connected to various actuators such as a ventilation fan, irrigation pump, and lighting system. Upon receiving control commands from the gateway, the actuator node activates or deactivates the corresponding devices to regulate the greenhouse environment. This enables automatic maintenance of ideal growing conditions without requiring continuous human intervention.

In the implemented system, the actuator node uses an ESP32 microcontroller connected to a relay module to control the ventilation fan, irrigation pump, and lighting system. The actuator receives control signals from the gateway through wireless communication and operates the devices accordingly to regulate airflow, temperature, lighting, and irrigation inside the greenhouse. The relay module acts as an electrical switching device, allowing the low-voltage signal from the ESP32 to safely control the power supplied to the actuators. This enables remote operation of the ventilation system, irrigation system, and lighting system through the dashboard interface.

The actuator node has been successfully expanded to include additional devices such as lighting systems and irrigation pumps, enabling complete control of greenhouse environmental conditions and demonstrating the effectiveness of the automated greenhouse control system.

III Applications and Future Scope:

The proposed smart greenhouse monitoring system has strong potential for real-world applications in greenhouse farming, urban gardening, and agricultural research, where continuous monitoring and control of environmental conditions are required. The system can be further enhanced by integrating additional sensors such as pH sensors, nutrient sensors, and advanced gas sensors to monitor more detailed soil and environmental conditions.

In future developments, machine learning algorithms can be integrated for predictive analysis and automatic decision-making based on historical environmental data. The addition of mobile applications, real-time alert and notification systems, and renewable energy sources such as solar power can further improve system efficiency, scalability, and sustainability. These improvements will enable the system to support large-scale agricultural environments and advanced precision farming techniques.

IV Result Photos:



Fig1: Image Showing Model of Greenhouse



Fig2: Image Showing Gateway with OLED Display

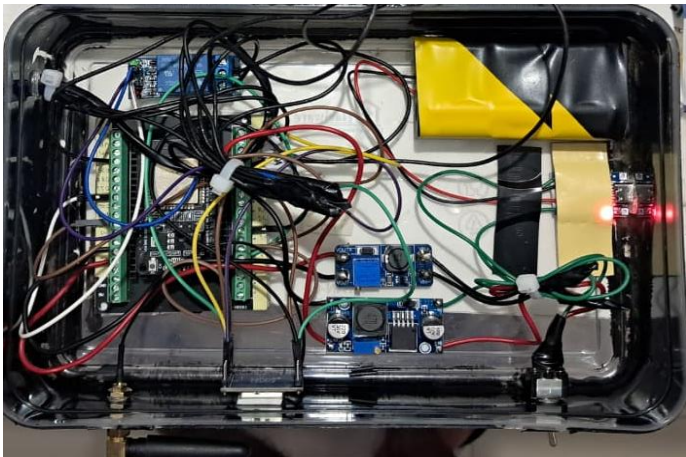


Fig3: Image showing the interior connections of Sensor Node

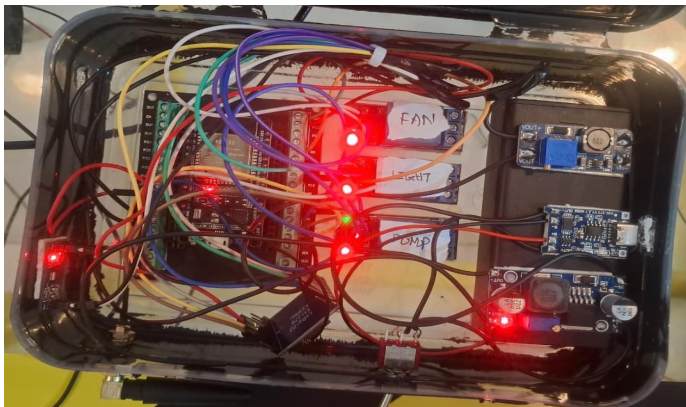


Fig4: Image showing the interior connections of Actuator Node

V References:

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- 8) Aqeel ur Rehman et al. 2022. Smart greenhouse monitoring system using IoT and wireless sensor networks. *International Journal of Advanced Computer Science and Applications* (2022).
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VI: Appendix – Datasheet References:

1. BH1750 Light Intensity Sensor

Description: Light Intensity Sensor

Key Specifications: Measurement Range: 1–65535 lux, Interface: I²C

Datasheet Link:

<https://rohmfs.rohm.com/en/products/databook/datasheet/ic/sensor/light/bh1750fvi-e.pdf>

2. Soil Moisture Sensor

Description: Soil Water Content Detection Sensor

Key Specifications: Voltage: 3.3–5V, Output Type: Analog

Datasheet Link: https://components101.com/sites/default/files/component_datasheet/Soil-Moisture-Sensor-Datasheet.pdf

3. SCD40 CO₂ Sensor

Description: Carbon Dioxide Sensor

Key Specifications: Measurement Range: 400–5000 ppm, Interface: I²C

Datasheet Link:

https://sensirion.com/media/documents/48C4B7FB/6165371E/Sensirion_CO2_Sensors_SCD4x_Datasheet.pdf

4. ESP32 Microcontroller

Description: Microcontroller Unit

Key Specifications: Dual Core Processor, Clock Frequency up to 240 MHz

Datasheet Link:

https://www.espressif.com/sites/default/files/documentation/esp32_datasheet_en.pdf

5. SX1278 LoRa Module

Description: LoRa Communication Module

Key Specifications: Operating Frequency: 433 MHz, Interface: SPI

Datasheet Link: https://www.semtech.com/uploads/documents/sx1276_77_78_79.pdf

6. Raspberry Pi Gateway

Description: Gateway Controller

Key Specifications: ARM Cortex Processor, Linux Operating System

Datasheet Link: <https://datasheets.raspberrypi.com/rpi4/raspberry-pi-4-datasheet.pdf>